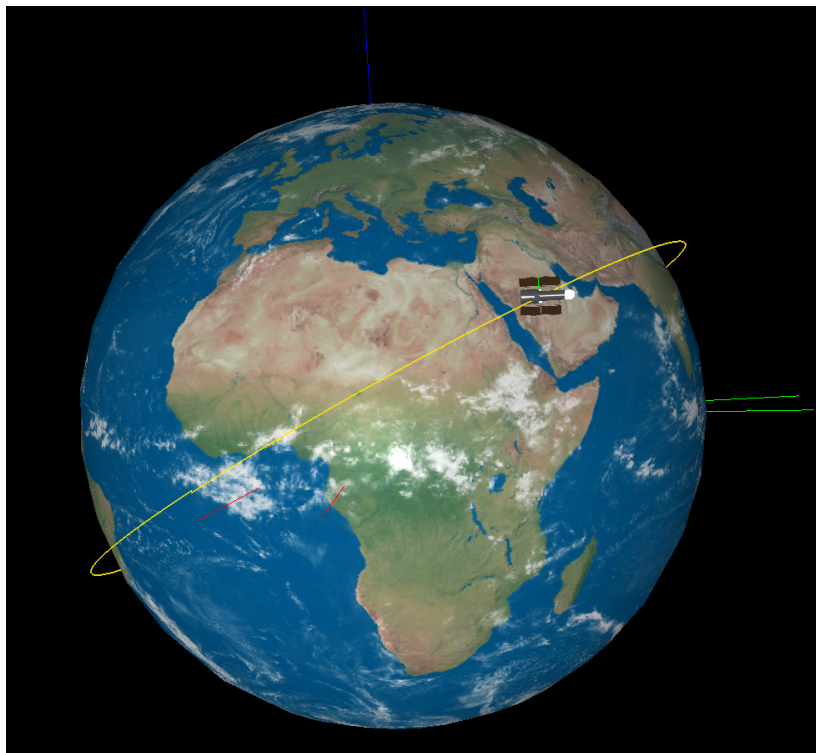


Hubble Space Telescope: Orbital and Attitude Simulation Study

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1 Modeling Framework and Assumptions

This study reuses the orbital and attitude libraries developed throughout the course. The following summarises the reference frames, models, and simplifying assumptions on which the results are based.

Reference frames and conventions. Four right-handed frames are used: the inertial ECI frame i ; the Earth-fixed ECEF frame obtained by rotating ECI through the Greenwich sidereal angle θ_G about $\hat{z}$; the orbital LVLH frame o built from the instantaneous position and velocity; and the spacecraft body frame b . Part 2 additionally uses the RSW/Gaussian frame G , related to LVLH by a 180° rotation about x [2]. Attitude is represented by unit quaternions in scalar-first convention, with q_{ib} mapping body to inertial vectors and $q_{XY} = q_{XZ} \otimes q_{ZY}$.

Orbit propagation (Part 1). The orbit is propagated with the PKepler model [3]. It includes the secular J_2 regression of the RAAN Ω , the precession of the argument of perigee ω , and the mean-motion drift from the TLE coefficients $\dot{n}$ and $\ddot{n}$. The model does not include atmospheric drag as a force (decay enters only through the frozen $\dot{n}$), higher-order gravity terms, or third-body and SRP perturbations. These omissions explain the divergence of long term decay predictions from reality.

Attitude dynamics and actuation (Part 2). The spacecraft is modelled as a rigid body with constant HST inertia $\mathbf{J}$. Reaction wheels and magnetic torquers are idealised as unity-gain actuators with per-axis saturation at ± 1.13 Nm, so wheel momentum build-up and magnetic momentum dumping are not represented. Disturbances include only the gravity-gradient torque and the prescribed thermal-snap disturbance $d(t)$.

Sensors and methods. The sensors are a three-axis rate gyro (white noise $\sigma_g^2 = 10^{-6}$, zero bias) and one or three star trackers using the assignment's axis-angle noise model ($\sigma^2 = 10^{-2}$). Gyro output is taken directly as ω_{ib}^b and star-tracker output as q_{ib} , so no attitude filter is used. Kepler's equation is solved iteratively, and quaternion kinematics are integrated with a fourth-order Runge-Kutta scheme. When multiple measurements are available (Task 2), Davenport's q -method and the TRIAD algorithm are used [1]. Control is performed by a feedback-linearised PD law and a computed-torque sliding-mode controller.

Mission context. HST currently orbits at ~ 475 km altitude and 28.47° inclination [4]. With no further reboost, NASA expects an uncontrolled reentry in the mid-to-late 2030s, with the estimate strongly modulated by solar activity [5].

AI Statement

Large language models such as Copilot and Gemini were used as coding assistants, since this course focuses on modelling rather than software engineering and my programming skills are not at the level required for a project of this size. They supported the workflow by helping debug Python scripts, interpret error messages, build advanced functions for simulation and logging, and compare long diagnostic outputs. The orbital and attitude models, the equations, and the implementation come from the course material and from the libraries I developed in previous assignments. All numerical results, figures, gains, and the final interpretation of the data presented in this report were independently verified and validated by me.

2 Part 1: Orbital Study

2.1 Task 1: Orbital parameters computation at epoch

Table 1 lists the values computed at the HST TLE epoch using the orbital functions of my custom libraries.

Table 1: Various orbital data and parameters at the HST TLE epoch.

Name	Symbol	Value
Specific Relative Angular Momentum	h	$5.2265 \times 10^4 \text{ km}^2/\text{s}$
True Anomaly	θ	4.889 rad
Eccentric Anomaly	E	4.889 rad
Semi-Major Axis	a	$6.853 \times 10^3 \text{ km}$
Mean Motion	n	$1.113 \times 10^{-3} \text{ rad/s}$
Derivative of Mean Motion	$\dot{n}$	$4.05 \times 10^{-14} \text{ rad/s}^2$
Second Derivative of Mean Motion	$\ddot{n}$	0
Position	$\mathbf{r}^i$	$[6.430 \times 10^3, -2.371 \times 10^3, 3.40] \text{ km}$
Velocity	$\mathbf{v}^i$	$[2.32, 6.29, 3.64] \text{ km/s}$
Julian Date	JD	2461170.7592
Sidereal Angle	θ_{G0}	5.600 rad
Orbit frame quaternion	$\mathbf{q}_{io}$	$[0.463, 0.662, 0.249, 0.535]$
Orbit frame Angular Velocity	$\boldsymbol{\omega}_{io}^i$	$[-1.84, -4.98, 9.78] \times 10^{-4} \text{ rad/s}$
Orbit frame Angular Acceleration	$\dot{\boldsymbol{\omega}}_{io}^i$	$[0, 0, 0] \text{ rad/s}^2$
Geodetic Latitude	λ'	$4.99 \times 10^{-4} \text{ rad}$
Geodetic Longitude	ϕ	0.330 rad
Altitude	h	$4.75 \times 10^2 \text{ km}$

2.2 Task 2: Long term decay and propagation accuracy

Orbital altitude 9 years into the future

Starting from the current HST TLE epoch, the altitude was integrated 9 years forward with the PKepler propagator. The result is shown in Figure 1.

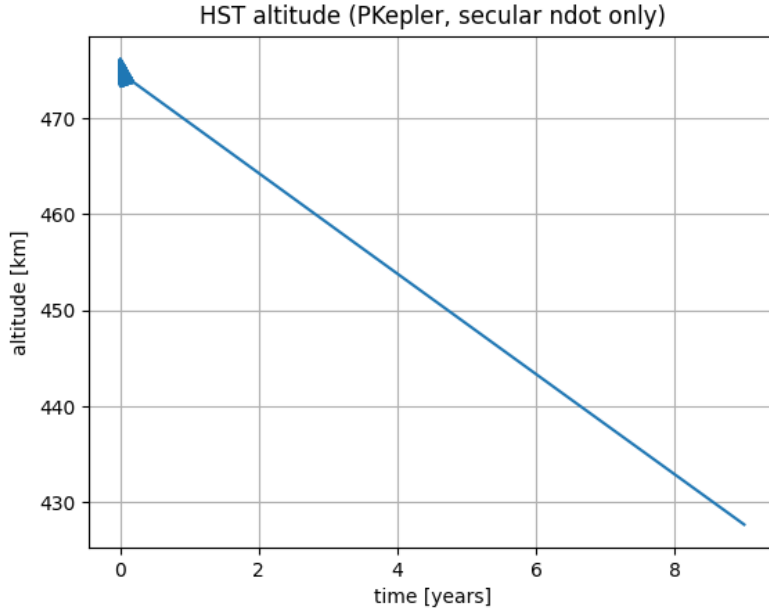


Figure 1: HST altitude predicted by PKepler over 9 years.

The altitude drops approximately linearly from

$$h_0 = 474.52 \text{ km} \longrightarrow h_{9\text{yr}} = 427.68 \text{ km},$$

a total decay of about 47 km over 9 years. The long term PKepler propagation does not reproduce NASA's estimate that HST will reenter in the mid-2030s. After nine years, the model still predicts an altitude far above the reentry threshold (120 km). This happens because PKepler uses a constant linear decay rate

based on the TLE's n , while real atmospheric drag is highly nonlinear. As the satellite descends, atmospheric density increases exponentially, which accelerates the decay. The real trajectory bends sharply downward in its final years, where PKepler simply extrapolates the initial decay rate as if it remained constant.

In general, obtaining accurate deorbit estimates is hard because they require; a coupled atmospheric drag model with a realistic density profile, a forecast of solar activity and to consider that heating expands the upper atmosphere and makes LEO drag inconsistent, the spacecraft's projected area and drag coefficient which are time dependent, and full numerical integration rather than analytic propagation.

Propagation error against an older TLE

To quantify the model error directly, the older HST TLE was propagated up to the epoch of the current TLE and compared against it. The propagation span is

$$3160.14 \text{ days} \approx 8.65 \text{ years}.$$

The resulting state vector error is extremely large:

$$\|\Delta r\| = 5393.76 \text{ km}, \quad \|\Delta v\| = 8.21 \text{ km/s}.$$

These values indicate that the propagated state is uncorrelated with the truth. The orbital element differences show that the shape parameters (a , e , i) remain close, while the phase parameters (Ω , ω) diverge.

$$\Delta a = +57.5 \text{ km}, \quad \Delta e = -2.20 \times 10^{-4}, \quad \Delta i = -0.0066^\circ,$$

$$\Delta \Omega = +187.59^\circ, \quad \Delta \omega = +280.07^\circ.$$

This behaviour is expected. In LEO, J drives continuous nodal regression and apsidal precession, and these effects accumulate over thousands of revolutions. Even small mismatches between the analytic PKepler rates and the real precession lead to large phase errors over multiple years. These phase errors dominate the long term state vector discrepancy and explain the huge position mismatch.

Short term behaviour and ground tracks

Over a single orbital period starting from the same epoch, PKepler and a two body Kepler propagation track each other almost identically: the position difference (Figure 2) increases by around 50 kilometres, and the ground tracks (Figure 3) are visually indistinguishable.

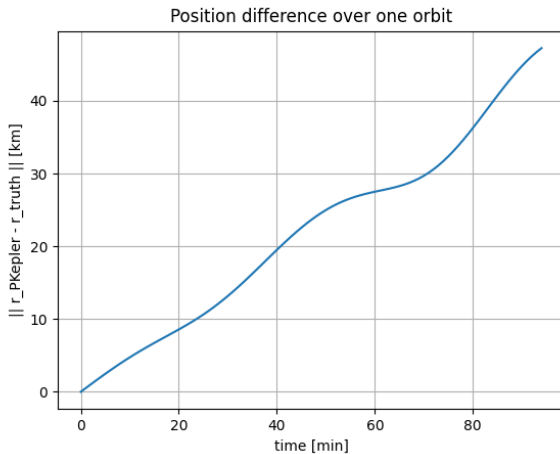


Figure 2: radius difference over one orbit

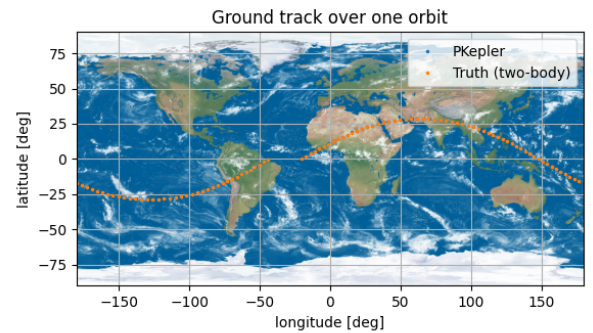


Figure 3: Ground tracks of PKepler and two body propagation over one orbit.

Possible improvements

The PKepler propagator is adequate for one orbit predictions but not for multiple year deorbit studies. Some of the upgrades could be:

- Use SGP4/SDP4 or a numerical integrator with a higher order gravity field and third body effects.
- Add an atmospheric drag model with a representative density and an estimate of the spacecraft's ballistic coefficient $C_D A/m$.
- Couple in a model of solar activity over the prediction horizon, which strongly affects drag in LEO and remove the assumption that the earth is transparent.
- Re-initialise from the freshest available TLE.

3 Part 2: Attitude Study

In Part 2 the Hubble Space Telescope is asked to hold the fixed inertial attitude

$$q_{id} = \frac{1}{2} [1 \ 1 \ 1 \ 1]^\top, \quad \omega_{id}^d = 0,$$

over four orbital periods, starting from the misaligned initial state $q_{ib}(0) = [1 \ 0 \ 0 \ 0]^\top$, $\omega_{ib}^b(0) = 10^{-3}[0.3, -0.1, 0.2]^\top$ rad/s. The body experiences gravity gradient torque, the solar array disturbance $d(t)$, actuator saturation, and noisy gyro and star-tracker measurements.

Setup

For the assignment, the actuator is treated as a unit gain rigid body torque with per-axis saturation at ± 1.13 Nm:

$$\tau_a = 1.13 \cdot \text{sat}(\tau_c/1.13), \quad \tau_{\text{total}} = \tau_a + \tau_g(r^i, q_{ib}, J) + \tau_d(t).$$

The solar-array disturbance is applied along the body y -axis, $\tau_d(t) = d(t) \hat{e}_y$, with

$$d(t) = A_1 \sin(p_1 t + \varphi_1) + A_2 \sin(p_2 t + \varphi_2),$$

and the assignment values $A_{1,2} = 0.2$ Nm, $p_1 = 0.14\pi$, $p_2 = 1.22\pi$, $\varphi_1 = 0.31\pi$, $\varphi_2 = -0.05\pi$.

The gyro noise is white, $\sigma_g^2 = 10^{-6}$ (so $\sigma_g \approx 10^{-3}$ rad/s) with zero bias; the star tracker model has $\sigma^2 = 10^{-2}$ (so $\sigma \approx 0.1$ rad $\approx 5.7^\circ$ per sample).

The attitude error quaternion is

$$q_{\text{err}} = q_{id}^{-1} \otimes q_{ib,\text{est}}, \quad q_{e,0} \geq 0,$$

and the pointing error in arc-seconds is computed from

$$e_\theta = \frac{2 \cdot 180 \cdot 3600}{\pi} \arcsin(\|q_{e,v}\|).$$

In addition to this measured error I log the true error $q_{id}^{-1} \otimes q_{ib,\text{true}}$, which is the physical pointing accuracy independent of measurement noise.

3.1 Task 1: PD baseline vs. Sliding-Mode Controller

Control laws

Both controllers use a feedback linearised structure

$$\tau_c = \omega \times J\omega + J u,$$

which cancels the rigid-body cross coupling and makes the closed loop dynamics approximately decoupled in the body frame, provided the actuators are not saturated. With this scaling, the assignment's PD gains generate torques of order $J k_1 \sim 9$ Nm/rad, which are the right magnitude for an HST-class actuator. They differ only in the feedback law u :

- **Baseline PD:** $u_{\text{PD}} = -k_1 q_v - k_2 \omega_{\text{err}}$ with the assignment gains $k_1 = 10^{-4}$, $k_2 = 2 \times 10^{-2}$.
- **Sliding-mode (SMC):** on the surface $s = \omega_{\text{err}} + 2k_1 q_v$ the controller applies

$$u_{\text{SM}} = -k_1 q_v - k \text{sat}(s/\varepsilon),$$

with tuned gains $k_1 = 3 \times 10^{-4}$, $k = 3 \times 10^{-3}$, $\varepsilon = 0.15$. The boundary layer ε is chosen so that within it the effective rate gain is $k/\varepsilon \approx 2 \times 10^{-2}$ – matched to PD’s k_2 – which keeps the controller from chattering against the gyro noise. Tuning was done by trial and error: smaller k_1 collapses to PD-like behaviour with no benefit; larger k_1 amplifies the very large star-tracker noise through the proportional term and degrades performance; a narrower ε pushes the controller into permanent bang-bang at the actuator limit.

Results

The simulation was run for 4 orbits with $\Delta t = 1$ s. Figure 4 compares the controllers on the same time axis; Figure 5 and Figure 6 show the individual error histories (both measured and true).

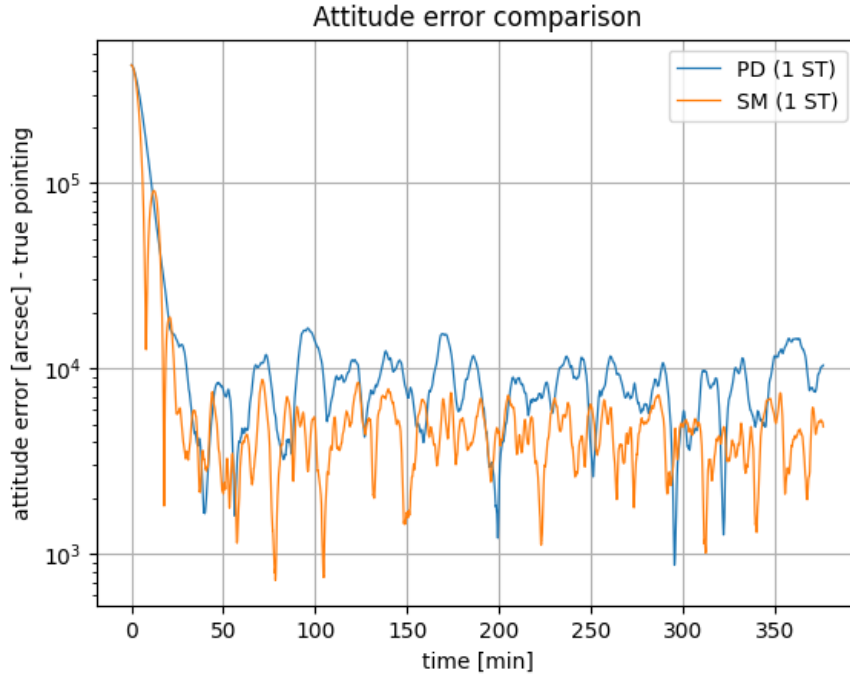


Figure 4: True pointing error vs. time for the baseline PD and the tuned SMC, both with one star tracker.

Averaged over the final orbit, the achieved pointing errors (true) are

$$\begin{aligned} \text{PD : } e_{\theta, \text{true}} &\approx 1.66 \times 10^4 \text{ arcsec } (\approx 4.61^\circ), \\ \text{SMC: } e_{\theta, \text{true}} &\approx 1.07 \times 10^4 \text{ arcsec } (\approx 2.97^\circ). \end{aligned}$$

Across multiple random seeds the SMC consistently reaches about *half* the RMS pointing error of the baseline PD ($\sim 1.24^\circ$ vs. $\sim 2.61^\circ$ in RMS terms).

Discussion

Both controllers achieve closed loop stability and bring HST from its $\approx 120^\circ$ initial misalignment to a steady oscillation around the desired inertial attitude. The SMC’s switching term gives noticeably better disturbance rejection: PD’s linear rate feedback responds to disturbances only in proportion to the resulting attitude error, where the SMC’s bang-bang term reacts strongly as soon as s leaves the boundary layer.

That said, the achieved pointing accuracy ($\sim 3^\circ$ at best) is many orders of magnitude away from HST’s flight specification of 0.007 arcsec. The dominant limitation in this simulation is the sensor noise: the star tracker, with $\sigma \approx 5.7^\circ$ per sample, enters the controller through q_v and produces noise torques of order

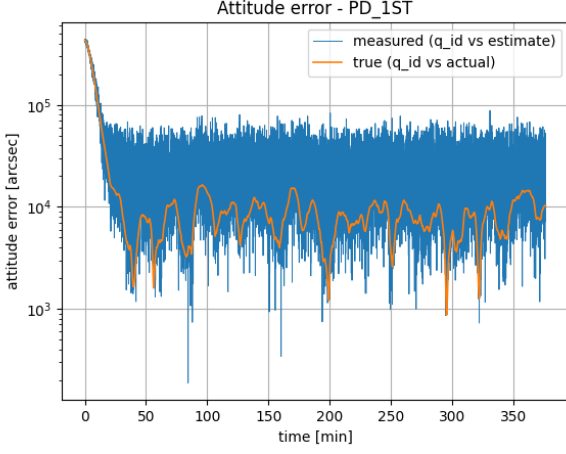


Figure 5: Baseline PD with one star tracker. The noisier trace is the *measured* error (limited by the $\sim 5.7^\circ$ ST noise); the smoother trace is the *true* error.

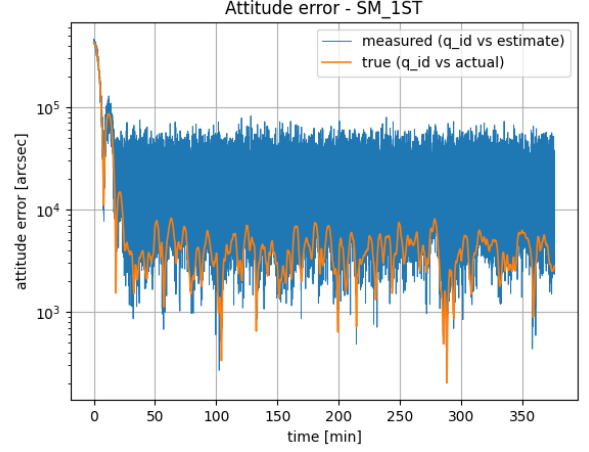


Figure 6: Tuned SMC with one star tracker.

$J \cdot k_1 \cdot \sigma$ that are comparable to the actuator limit. Increasing the gains amplifies this rather than suppressing it. The real HST achieves its specification with much finer Fine Guidance Sensors, four redundancy gyros, and proper attitude filtering (Kalman style).

3.2 Task 2: One star tracker vs. three (Davenport averaging)

To evaluate whether multiple star trackers improve the attitude estimate, two additional trackers were added to the simulation. Each tracker produces an independent noisy quaternion measurement following the same axis-angle noise model used in Task 1 ($\sigma^2 = 10^{-2}$). The three measurements q_1, q_2, q_3 are fused using Davenport’s q -method. For each tracker i , the reference basis vectors (a_i, b_i) are rotated through the measured quaternion to obtain the corresponding observed vectors $(\hat{a}_i, \hat{b}_i)$. These six vector pairs are then used to build Davenport’s K -matrix, where the dominant eigenvector provides the averaged attitude quaternion.

This procedure is implemented in the `DavenportMultiST` class of `sat_lib.9.py`. The resulting fused attitude estimate is passed directly to the same SMC controller used in Task 1.

Averaged over the final orbit, the results are :

$$\begin{aligned} 1 \text{ ST: } e_{\text{meas,mean}} &\approx 2.33 \times 10^4 \text{ arcsec,} \\ 3 \text{ ST: } e_{\text{meas,mean}} &\approx 1.80 \times 10^4 \text{ arcsec,} \end{aligned}$$

i.e. a reduction of about 23% in the measured error, in good agreement with the theoretical $\sqrt{3}$ noise-reduction factor of averaging three independent estimates.

True vs. measured pointing

A more subtle observation – and the one that I think actually drives the discussion of this task – is that the *true* pointing accuracy barely changes:

$$e_{\text{true,mean}} \approx 1.04 \times 10^4 \text{ arcsec } (\approx 2.9^\circ)$$

for both the 1-ST and 3-ST configurations. The reason could be that HST’s very large inertia ($J \sim 10^5 \text{ kg m}^2$) acts as a strong low pass filter on the high-frequency measurement noise: the closed-loop response cannot move the spacecraft fast enough to chase the noise torque, so even though the controller sees a less noisy estimate with three trackers, the spacecraft itself does not actually point appreciably better. What ultimately limits the true pointing accuracy here is not the star tracker noise but the combination of the periodic solar-array disturbance, the gravity gradient torque and the limited control bandwidth allowed by actuator saturation.

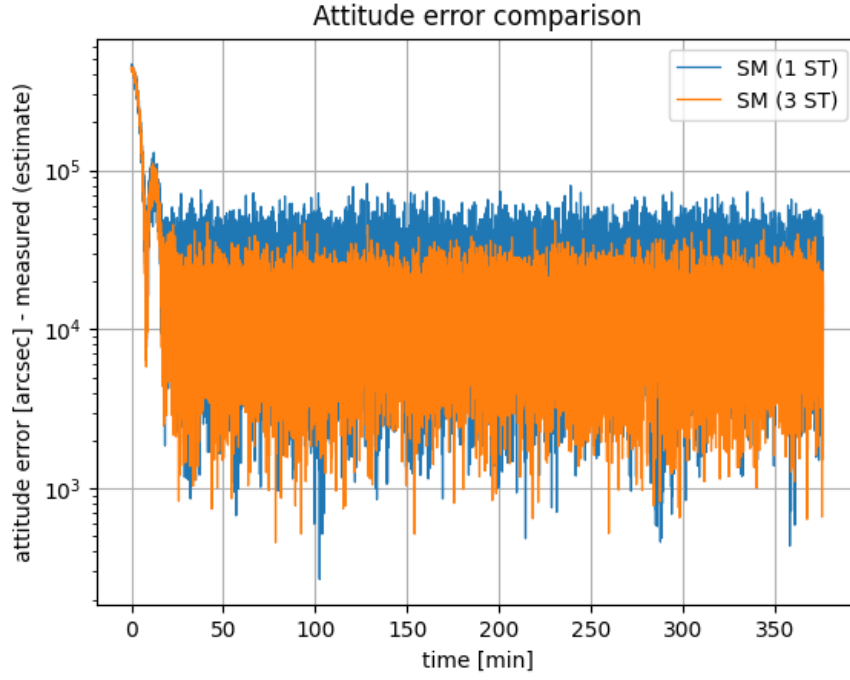


Figure 7: Measured attitude error vs. time for the SMC with one and three star trackers.

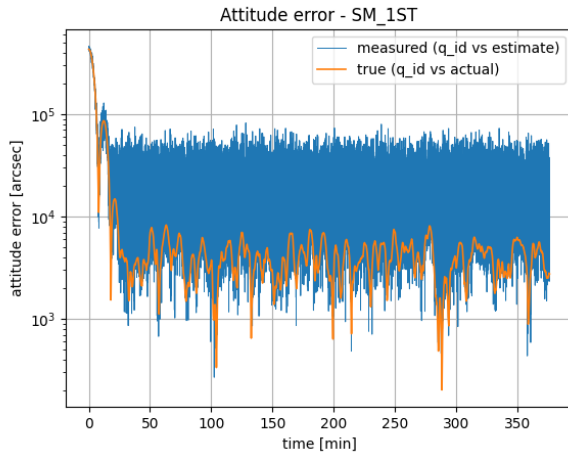


Figure 8: SMC with one star tracker.

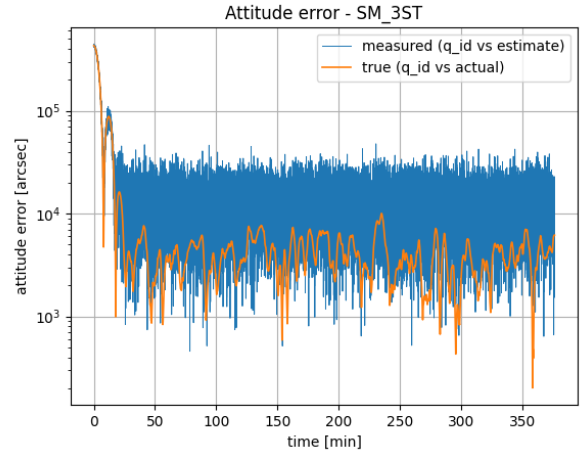


Figure 9: SMC with three star trackers and Davenport averaging.

This matches the physical situation on the real HST as well: multiple star trackers are flown primarily for redundancy/fault tolerance and to enlarge the available sky coverage; the ultimate pointing accuracy is set by the Fine Guidance Sensors, the reaction wheel control performance and the disturbance suppression hardware.

Animation

A 3D animation was generated to visualise the attitude motion from Task 2. The animation shows the spacecraft in low Earth orbit with its body frame and the ECI frame rendered, together with the Keplerian reference orbit traced in inertial space.

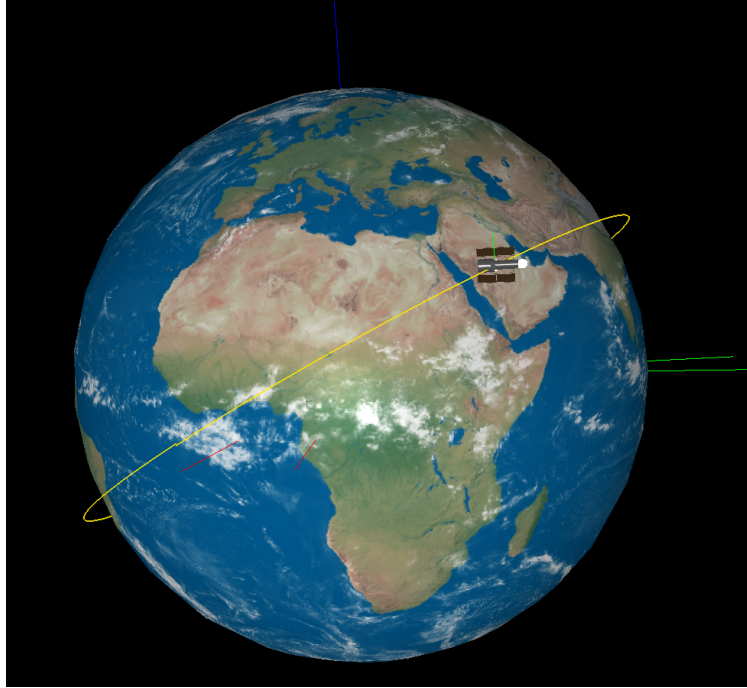


Figure 10: Representative frame from the Task 2 animation.

Conclusion. Overall, the attitude study shows that both controllers achieve stable tracking under realistic disturbances, but with different performance. The SMC provides stronger disturbance rejection and consistently lower true pointing error than the baseline PD, although both remain far from HST’s flight accuracy due to the coarse sensor noise and actuator limits imposed by the assignment. The multi-star tracker configuration reduces the measurement noise by the expected $\sqrt{3}$ factor, yet the true pointing accuracy remains basically unchanged because the spacecraft’s large inertia and limited control bandwidth filter out high frequency variations. In practice, this means that improving the estimator alone is not enough: achieving subarcsecond performance requires finer sensors, higher bandwidth actuation, and proper attitude filtering.

References

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